

Data Visualization

The purpose of this notebook is to practice data visualization, and hopefully communicate some best-practices along the way.

Some sources I'd reccomend for data visualization principles are:

- Storytelling with data, C. Knaflie
- The visual display of quantitative information, E. Tufte
- Better data visualizations, J. Schwabish

I have other notebooks that also incoroprate some nice visuals:

UK COVID-19 Vaccination Data Visualization

Exploratory Data Visualization - Student Performance

Netflix Data Visualization Project

HR Data Set - Visuals & Predictions

Visuals & Customer Segmentation

With that said, let's go...

```
[1]: # This Python 3 environment comes with many helpful analytics libraries installed

# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)


import os

for filename in filenames:
    print(os.path.join(dirname, filename))

# You can also write temporary files to , but they won't be saved outside of the current session
```

```
[2]: # Importing libs
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AffinityPropagation
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
import warnings
warnings.filterwarnings("ignore")
import plotly as py
import plotly.graph_objs as go
import os
py.offline.init_notebook_mode(connected = True)
#print(os.listdir("../input"))
import datetime as dt
import missingno as msno
plt.rcParams['figure.dpi'] = 140
```

```
[3]: df.head(3)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	desc
0	s1	TV Show	3%	NaN	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	August 14, 2020	2020	TV-MA	4 Seasons	International TV Shows, TV Dramas, TV Sci-Fi &...	In a future where the inhabit an island ...
1	s2	Movie	7:19	Jorge Michel Grau	Demián Bichir, Héctor Bonilla, Oscar Serrano, ...	Mexico	December 23, 2016	2016	TV-MA	93 min	Dramas, International Movies	After a devastating earthquake hits Mexi Cit...

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	desc
2	s3	Movie	23:59	Gilbert Chan	Tedd Chan, Stella Chung, Henley Hii, Lawrence ...	Singapore	December 20, 2018	2011	R	78 min	Horror Movies, International Movies	When an army recruit found dead, his fellow

```
[4]: # Missing data

for i in df.columns:
    null_rate = df[i].isna().sum() / len(df) * 100
    if null_rate > 0 :
        print("{} null rate: {}".format(i,round(null_rate,2)))
```

```
director null rate: 30.68%
cast null rate: 9.22%
country null rate: 6.51%
date_added null rate: 0.13%
rating null rate: 0.09%
```

- 5 columns have missing values, with Director missing 1/3 of the time

Dealing with the missing data

- This is always scenario dependant, but in this case, I will:
 - replace blank countries with the mode (most common) country
 - I want to keep director as it could be interesting to look at a certain director's films
 - I want to keep cast as it could be interesting to look at a certain cast's films

```
[5]: # Replacments

df['country'] = df['country'].fillna(df['country'].mode()[0])

df['cast'].replace(np.nan, 'No Data',inplace = True)
df['director'].replace(np.nan, 'No Data',inplace = True)

# Drops

df.dropna(inplace=True)

# Drop Duplicates

df.drop_duplicates(inplace= True)
```

```
[6]: df.isnull().sum()
```

```
[6]: show_id      0
type          0
title         0
director      0
cast          0
country       0
date_added    0
release_year  0
rating        0
duration      0
listed_in     0
description   0
dtype: int64
```

```
[7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 7770 entries, 0 to 7786
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   show_id         7770 non-null   object
1   type            7770 non-null   object
2   title           7770 non-null   object
3   director        7770 non-null   object
4   cast            7770 non-null   object
5   country         7770 non-null   object
6   date_added      7770 non-null   object
7   release_year    7770 non-null   int64
8   rating          7770 non-null   object
9   duration        7770 non-null   object
10  listed_in       7770 non-null   object
11  description      7770 non-null   object
dtypes: int64(1), object(11)
memory usage: 789.1+ KB
```

Missing values dealt with, but the date isn't quite right yet...

```
[8]: df["date_added"] = pd.to_datetime(df['date_added'])

df['month_added']=df['date_added'].dt.month
df['month_name_added']=df['date_added'].dt.month_name()
df['year_added'] = df['date_added'].dt.year

df.head(3)
```

```
[8]:
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	month_added	month_name_added	year
0	s1	TV Show	3%	No Data	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	2020-08-14	2020	TV-MA	4 Seasons	International TV Shows, TV Dramas, TV Sci-Fi &...	In a future where the elite inhabit an island ...	8	August	2020
1	s2	Movie	7:19	Jorge Michel Grau	Demián Bichir, Héctor Bonilla, Oscar Serrano, ...	Mexico	2016-12-23	2016	TV-MA	93 min	Dramas, International Movies	After a devastating earthquake hits Mexico Cit...	12	December	2016
2	s3	Movie	23:59	Gilbert Chan	Tedd Chan, Stella Chung, Henley Hii, Lawrence ...	Singapore	2018-12-20	2011	R	78 min	Horror Movies, International Movies	When an army recruit is found dead, his fellow...	12	December	2018

Okay, let's visualize

Where possible, I'll use the Netflix brand colours

<https://brand.netflix.com/en/assets/brand-symbol/>

Using a consistent color palette is a great way to give your work credibility. It looks professional, and keeps the reader engaged.

It's an easy-to-implement tip that really helps.

```
[9]:
```

```
# Palette
sns.palplot(['#221f1f', '#b20710', '#e50914', '#f5f5f1'])

plt.title("Netflix brand palette ",loc='left',fontfamily='serif',fontsize=15,y=1.2)
plt.show()
```

Netflix brand palette



Netflix through the years

Netflix started as DVD rentals, and now they have an audience of over 150m people - this is their story...

```
[10]:
```

```
from datetime import datetime

## these go on the numbers below
tl_dates = [
    "1997\nFounded",
    "1998\nMail Service",
    "2003\nGoes Public",
    "2007\nStreaming service",
    "2016\nGoes Global",
    "2021\nNetflix & Chill"
]

tl_x = [1, 2, 4, 5.3, 8,9]

## these go on the numbers
tl_sub_x = [1.5,3,5,6.5,7]

tl_sub_times = [
    "1998", "2000", "2006", "2010", "2012"
]

tl_text = [
    "Netflix.com launched",
    "Starts\nPersonal\nRecommendations", "Billionth DVD Delivery", "Canadian\nLaunch", "UK Launch\n(my birthplace)"]
```

```

# Set figure & Axes
fig, ax = plt.subplots(figsize=(15, 4), constrained_layout=True)
ax.set_ylim(-2, 1.75)
ax.set_xlim(0, 10)

# Timeline : line
ax.axhline(0, xmin=0.1, xmax=0.9, c='#4a4a4a', zorder=1)

# Timeline : Date Points
ax.scatter(tl_x, np.zeros(len(tl_x)), s=120, c='#4a4a4a', zorder=2)
ax.scatter(tl_x, np.zeros(len(tl_x)), s=30, c='#fafafa', zorder=3)
# Timeline : Time Points
ax.scatter(tl_sub_x, np.zeros(len(tl_sub_x)), s=50, c='#4a4a4a', zorder=4)

# Date Text
for x, date in zip(tl_x, tl_dates):
    ax.text(x, -0.55, date, ha='center',
            fontfamily='serif', fontweight='bold',
            color='#4a4a4a', fontsize=12)

# Stemplot : vertical line
levels = np.zeros(len(tl_sub_x))
levels[::2] = 0.3
levels[1::2] = -0.3
markerline, stemline, baseline = ax.stem(tl_sub_x, levels, use_line_collection=True)
plt.setp(baseline, zorder=0)
plt.setp(markerline, marker=',', color='#4a4a4a')
plt.setp(stemline, color='#4a4a4a')

# Text
for idx, x, time, txt in zip(range(1, len(tl_sub_x)+1), tl_sub_x, tl_sub_times, tl_text):
    ax.text(x, 1.3*(idx%2)-0.5, time, ha='center',
            fontfamily='serif', fontweight='bold',
            color='#4a4a4a' if idx!=len(tl_sub_x) else '#b20710', fontsize=11)

    ax.text(x, 1.3*(idx%2)-0.6, txt, va='top', ha='center',
            fontfamily='serif', color='#4a4a4a' if idx!=len(tl_sub_x) else '#b20710')

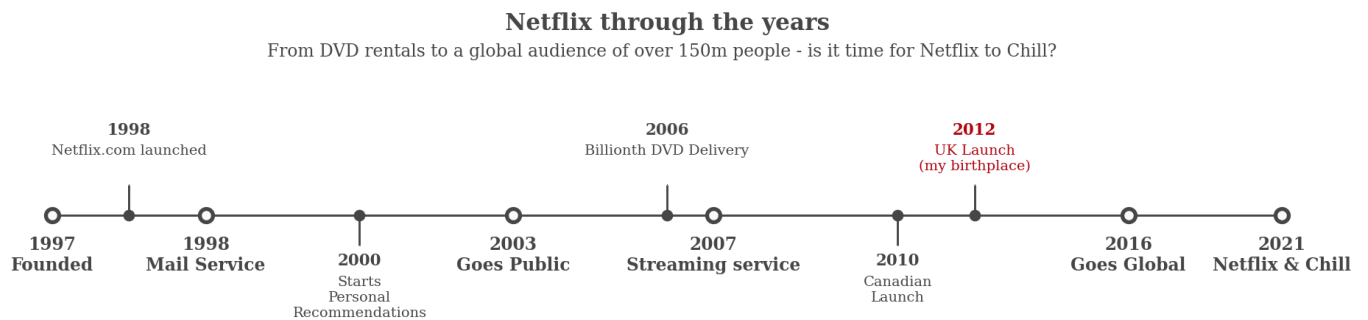
# Spine
for spine in ["left", "top", "right", "bottom"]:
    ax.spines[spine].set_visible(False)

# Ticks
ax.set_xticks([])
ax.set_yticks([])

# Title
ax.set_title("Netflix through the years", fontweight="bold", fontfamily='serif', fontsize=16, color='#4a4a4a')
ax.text(2.4, 1.57, "From DVD rentals to a global audience of over 150m people - is it time for Netflix to Chill?", fontfamily='serif', fontsize=12, color='#4a4a4a')

plt.show()

```



Content - Let's explore

Now we've seen how Netflix came to dominate our TV screens, let's have a look at the content they offer...

[11]: # For viz: Ratio of Movies & TV shows

```

x=df.groupby(['type'])['type'].count()
y=len(df)
r=((x/y)).round(2)

mf_ratio = pd.DataFrame(r).T

```

```

[12]: fig, ax = plt.subplots(1,1,figsize=(6.5, 2.5))

ax.barh(mf_ratio.index, mf_ratio['Movie'],
        color='#b20710', alpha=0.9, label='Male')
ax.barh(mf_ratio.index, mf_ratio['TV Show'], left=mf_ratio['Movie'],
        color='#221f1f', alpha=0.9, label='Female')

ax.set_xlim(0, 1)
ax.set_xticks([])
ax.set_yticks([])
#ax.set_yticklabels(mf_ratio.index, fontfamily='serif', fontsize=11)

# movie percentage
for i in mf_ratio.index:
    ax.annotate(f"{int(mf_ratio['Movie'][i]*100)}%",
                xy=(mf_ratio['Movie'][i]/2, i),
                va = 'center', fontsize=40, fontweight='light', fontfamily='serif',
                color='white')

    ax.annotate("Movie",
                xy=(mf_ratio['Movie'][i]/2, -0.25),
                va = 'center', ha='center', fontsize=15, fontweight='light', fontfamily='serif',
                color='white')

for i in mf_ratio.index:
    ax.annotate(f"{int(mf_ratio['TV Show'][i]*100)}%",
                xy=(mf_ratio['Movie'][i]+mf_ratio['TV Show'][i]/2, i),
                va = 'center', ha='center', fontsize=40, fontweight='light', fontfamily='serif',
                color='white')
    ax.annotate("TV Show",
                xy=(mf_ratio['Movie'][i]+mf_ratio['TV Show'][i]/2, -0.25),
                va = 'center', ha='center', fontsize=15, fontweight='light', fontfamily='serif',
                color='white')

# Title & Subtitle
fig.text(0.125,1.03,'Movie & TV Show distribution', fontfamily='serif',fontsize=15, fontweight='bold')
fig.text(0.125,0.92,'We see vastly more movies than TV shows on Netflix.',fontfamily='serif',fontsize=12)

for s in ['top', 'left', 'right', 'bottom']:
    ax.spines[s].set_visible(False)

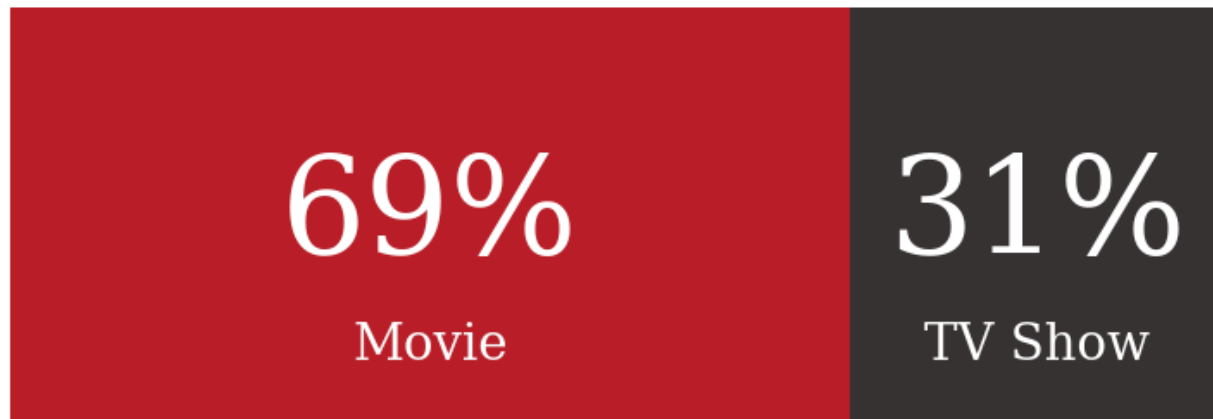
#ax.legend(loc='lower center', ncol=3, bbox_to_anchor=(0.5, -0.06))

# Removing legend due to labelled plot
ax.legend().set_visible(False)
plt.show()

```

Movie & TV Show distribution

We see vastly more movies than TV shows on Netflix.



By Country

So we now know there are much more movies than TV shows on Netflix (which surprises me!).

What about if we look at content by country?

I would imagine that the USA will have the most content. I wonder how my country, the UK, will compare?

```
[13]: # Quick feature engineering

# Helper column for various plots
df['count'] = 1

# Many productions have several countries listed - this will skew our results , we'll grab the first one mentioned

# Lets retrieve just the first country
df['first_country'] = df['country'].apply(lambda x: x.split(",")[0])
df['first_country'].head()

# Empty box

ratings_ages = {
    'TV-PG': 'Older Kids',
    'TV-MA': 'Adults',
    'TV-Y7-FV': 'Older Kids',
    'TV-Y7': 'Older Kids',
    'TV-14': 'Teens',
    'R': 'Adults',
    'TV-Y': 'Kids',
    'NR': 'Adults',
    'PG-13': 'Teens',
    'TV-G': 'Kids',
    'PG': 'Older Kids',
    'G': 'Kids',
    'UR': 'Adults',
    'NC-17': 'Adults'
}

df['target_ages'] = df['rating'].replace(ratings_ages)
df['target_ages'].unique()

# Genre

df['genre'] = df['listed_in'].apply(lambda x : x.replace(' ','').replace(', ','').split(','))

# Reducing name length

df['first_country'].replace('United States', 'USA', inplace=True)
df['first_country'].replace('United Kingdom', 'UK', inplace=True)
df['first_country'].replace('South Korea', 'S. Korea', inplace=True)
```

```
[14]: data = df.groupby('first_country')['count'].sum().sort_values(ascending=False)[:10]

# Plot
```

```

color_map = ['#f5f5f1' for _ in range(10)]
color_map[0] = color_map[1] = color_map[2] = '#b20710' # color highlight

fig, ax = plt.subplots(1,1, figsize=(12, 6))
ax.bar(data.index, data, width=0.5,
       edgecolor='darkgray',
       linewidth=0.6,color=color_map)

#annotations
for i in data.index:
    ax.annotate(f"{data[i]}",
               xy=(i, data[i] + 150), #i like to change this to roughly 5% of the highest cat
               va = 'center', ha='center',fontweight='light', fontfamily='serif')

# Remove border from plot
for s in ['top', 'left', 'right']:
    ax.spines[s].set_visible(False)

# Tick labels
ax.set_xticklabels(data.index, fontfamily='serif', rotation=0)

# Title and sub-title
fig.text(0.09, 1, 'Top 10 countries on Netflix', fontsize=15, fontweight='bold', fontfamily='serif')
fig.text(0.09, 0.95, 'The three most frequent countries have been highlighted.', fontsize=12, fontweight='light', fontfamily='serif')

fig.text(1.1, 1.01, 'Insight', fontsize=15, fontweight='bold', fontfamily='serif')

fig.text(1.1, 0.67, '''
The most prolific producers of
content for Netflix are, primarily,
the USA, with India and the UK
a significant distance behind.

It makes sense that the USA produces
the most content as, afterall,
Netflix is a US company.
''',
        , fontsize=12, fontweight='light', fontfamily='serif')

ax.grid(axis='y', linestyle='-', alpha=0.4)

grid_y_ticks = np.arange(0, 4000, 500) # y ticks, min, max, then step
ax.set_yticks(grid_y_ticks)
ax.set_axisbelow(True)

#Axis labels
#plt.xlabel("Country", fontsize=12, fontweight='light', fontfamily='serif',loc='left',y=-1.5)
#plt.ylabel("Count", fontsize=12, fontweight='light', fontfamily='serif')
#plt.legend(loc='upper right')

# thicken the bottom line if you want to
plt.axhline(y = 0, color = 'black', linewidth = 1.3, alpha = .7)

ax.tick_params(axis='both', which='major', labels=12)

import matplotlib.lines as lines
l1 = lines.Line2D([1, 1], [0, 1], transform=fig.transFigure, figure=fig,color='black',lw=0.2)
fig.lines.extend([l1])

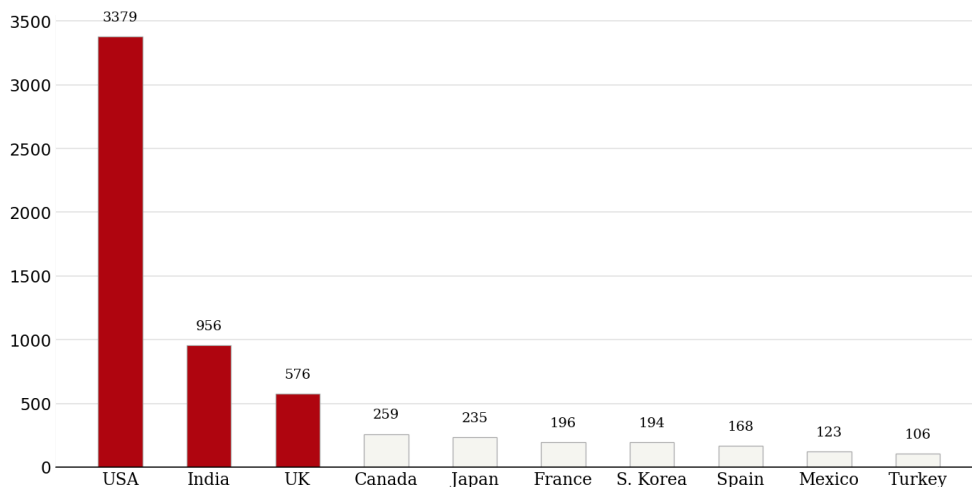
ax.tick_params(axis=u'both', which=u'both',length=0)

plt.show()

```

Top 10 countries on Netflix

The three most frequent countries have been highlighted.



Insight

The most prolific producers of content for Netflix are, primarily the USA, with India and the UK a significant distance behind.

It makes sense that the USA produces the most content as, after all, Netflix is a US company.

As predicted, the USA dominates.

The UK is a top contender too, but still some way behind India.

How does content by country vary?

```
[15]: country_order = df['first_country'].value_counts()[1:].index
data_q2q3 = df[['type', 'first_country']].groupby('first_country')['type'].value_counts().unstack().loc[country_order]
data_q2q3['sum'] = data_q2q3.sum(axis=1)
data_q2q3_ratio = (data_q2q3.T / data_q2q3['sum']).T[['Movie', 'TV Show']].sort_values(by='Movie', ascending=False)[::-1]

###
fig, ax = plt.subplots(1,1,figsize=(15, 8),)

ax.barh(data_q2q3_ratio.index, data_q2q3_ratio['Movie'],
        color='#b20710', alpha=0.8, label='Movie')
ax.barh(data_q2q3_ratio.index, data_q2q3_ratio['TV Show'], left=data_q2q3_ratio['Movie'],
        color='#221f1f', alpha=0.8, label='TV Show')

ax.set_xlim(0, 1)
ax.set_xticks([])
ax.set_yticklabels(data_q2q3_ratio.index, fontfamily='serif', fontsize=11)

# male percentage
for i in data_q2q3_ratio.index:
    ax.annotate(f"{data_q2q3_ratio['Movie'][i]*100:.3}%",
               xy=(data_q2q3_ratio['Movie'][i]/2, i),
               va = 'center', ha='center', fontsize=12, fontweight='light', fontfamily='serif',
               color='white')

for i in data_q2q3_ratio.index:
    ax.annotate(f"{data_q2q3_ratio['TV Show'][i]*100:.3}%",
               xy=(data_q2q3_ratio['Movie'][i]+data_q2q3_ratio['TV Show'][i]/2, i),
               va = 'center', ha='center', fontsize=12, fontweight='light', fontfamily='serif',
               color='white')

fig.text(0.13, 0.93, 'Top 10 countries Movie & TV Show split', fontsize=15, fontweight='bold', fontfamily='serif')
fig.text(0.131, 0.89, 'Percent Stacked Bar Chart', fontsize=12, fontfamily='serif')

for s in ['top', 'left', 'right', 'bottom']:
    ax.spines[s].set_visible(False)

#ax.legend(loc='lower center', ncol=3, bbox_to_anchor=(0.5, -0.06))

fig.text(0.75,0.9,"Movie", fontweight="bold", fontfamily='serif', fontsize=15, color='#b20710')
fig.text(0.81,0.9,"|", fontweight="bold", fontfamily='serif', fontsize=15, color='black')
fig.text(0.82,0.9,"TV Show", fontweight="bold", fontfamily='serif', fontsize=15, color='#221f1f')

fig.text(1.1, 0.93, 'Insight', fontsize=15, fontweight='bold', fontfamily='serif')

fig.text(1.1, 0.44, '''
Interestingly, Netflix in India
```

```
is made up nearly entirely of Movies.

Bollywood is big business, and perhaps
the main focus of this industry is Movies
and not TV Shows.

South Korean Netflix on the other hand is
almost entirely TV Shows.

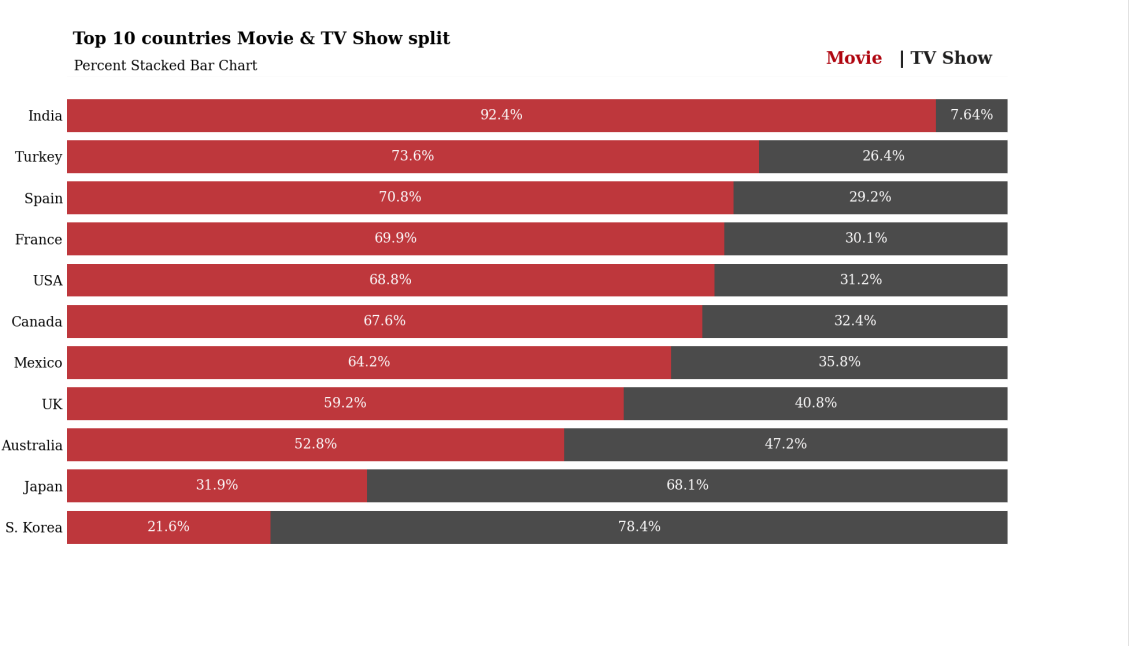
The underlying reasons for the difference
in content must be due to market research
conducted by Netflix.
'''

, fontsize=12, fontweight='light', fontfamily='serif')

import matplotlib.lines as lines
l1 = lines.Line2D([1, 1], [0, 1], transform=fig.transFigure, figure=fig,color='black',lw=0.2)
fig.lines.extend([l1])

ax.tick_params(axis='both', which='major', labelsize=12)
ax.tick_params(axis=u'both', which=u'both',length=0)

plt.show()
```



Insight

Interestingly, Netflix in India is made up nearly entirely of Movie

Bollywood is big business, and per the main focus of this industry is M and not TV Shows.

South Korean Netflix on the other h almost entirely TV Shows.

The underlying reasons for the difference in content must be due to market research conducted by Netflix.

As I've noted in the insights on the plot, it is really interesting to see how the split of TV Shows and Movies varies by country.

South Korea is dominated by TV Shows - why is this? I am a huge fan of South Korean cinema so I know they have a great movie selection.

Equally, India is dominated by Movies. I think this might be due to Bollywood - comment below if you have any other ideas!

Ratings

Let's briefly check out how ratings are distributed

```
[16]: order = pd.DataFrame(df.groupby('rating')['count'].sum().sort_values(ascending=False).reset_index())
rating_order = list(order['rating'])

[17]: mf = df.groupby('type')['rating'].value_counts().unstack().sort_index().fillna(0).astype(int)[rating_order]

movie = mf.loc['Movie']
tv = - mf.loc['TV Show']

fig, ax = plt.subplots(1,1, figsize=(12, 6))
ax.bar(movie.index, movie, width=0.5, color='#b20710', alpha=0.8, label='Movie')
ax.bar(tv.index, tv, width=0.5, color='#221f1f', alpha=0.8, label='TV Show')
#ax.set_ylim(-35, 50)

# Annotations
for i in tv.index:
    ax.annotate(f"~-tv[{i}]",
```

```

xy=(i, tv[i] - 60),
va = 'center', ha='center',fontweight='light', fontfamily='serif',
color='#4a4a4a')

for i in movie.index:
    ax.annotate(f"{movie[i]}",
               xy=(i, movie[i] + 60),
               va = 'center', ha='center',fontweight='light', fontfamily='serif',
               color='#4a4a4a')

for s in ['top', 'left', 'right', 'bottom']:
    ax.spines[s].set_visible(False)

ax.set_xticklabels(mf.columns, fontfamily='serif')
ax.set_yticks([])

ax.legend().set_visible(False)
fig.text(0.16, 1, 'Rating distribution by Film & TV Show', fontsize=15, fontweight='bold', fontfamily='serif')
fig.text(0.16, 0.89,
        '''We observe that some ratings are only applicable to Movies.
        The most common for both Movies & TV Shows are TV-MA and TV-14.
        ''',
        , fontsize=12, fontweight='light', fontfamily='serif')

fig.text(0.755,0.924,"Movie", fontweight="bold", fontfamily='serif', fontsize=15, color='#b20710')
fig.text(0.815,0.924,"|", fontweight="bold", fontfamily='serif', fontsize=15, color='black')
fig.text(0.825,0.924,"TV Show", fontweight="bold", fontfamily='serif', fontsize=15, color='#221f1f')

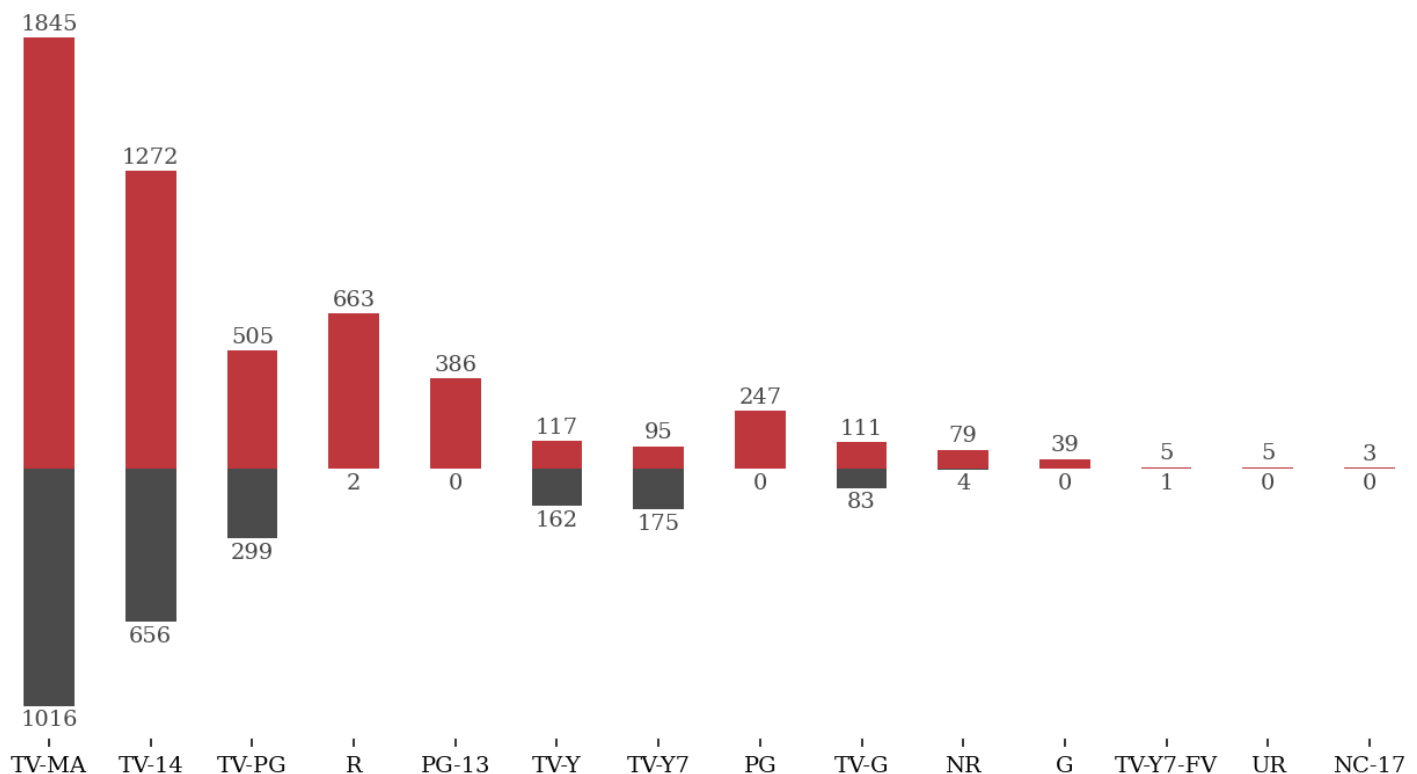
plt.show()

```

Rating distribution by Film & TV Show

We observe that some ratings are only applicable to Movies.
The most common for both Movies & TV Shows are TV-MA and TV-14.

Movie | **TV Show**



How has content been added over the years?

As we saw in the timeline at the start of this analysis, Netflix went global in 2016 - and it is extremely noticeable in this plot.

The increase in Movie content is remarkable.

[18]:

```
fig, ax = plt.subplots(1, 1, figsize=(12, 6))
```

```

color = ["#b20710", "#221f1f"]

for i, mtv in enumerate(df['type'].value_counts().index):
    mtv_rel = df[df['type']==mtv]['year_added'].value_counts().sort_index()
    ax.plot(mtv_rel.index, mtv_rel, color=color[i], label=mtv)
    ax.fill_between(mtv_rel.index, 0, mtv_rel, color=color[i], alpha=0.9)

ax.yaxis.tick_right()

ax.axhline(y = 0, color = 'black', linewidth = 1.3, alpha = .7)

#ax.set_ylim(0, 50)
#ax.legend(loc='upper left')
for s in ['top', 'right', 'bottom', 'left']:
    ax.spines[s].set_visible(False)

ax.grid(False)

ax.set_xlim(2008, 2020)
plt.xticks(np.arange(2008, 2021, 1))

fig.text(0.13, 0.85, 'Movies & TV Shows added over time', fontsize=15, fontweight='bold', fontfamily='serif')
fig.text(0.13, 0.59,
'''We see a slow start for Netflix over several years.
Things begin to pick up in 2015 and then there is a
rapid increase from 2016.

It looks like content additions have slowed down in 2020,
likely due to the COVID-19 pandemic.
''',
, fontsize=12, fontweight='light', fontfamily='serif')

fig.text(0.13, 0.2, "Movie", fontweight="bold", fontfamily='serif', fontsize=15, color='#b20710')
fig.text(0.19, 0.2, "|", fontweight="bold", fontfamily='serif', fontsize=15, color='black')
fig.text(0.2, 0.2, "TV Show", fontweight="bold", fontfamily='serif', fontsize=15, color='#221f1f')

ax.tick_params(axis=u'both', which=u'both',length=0)

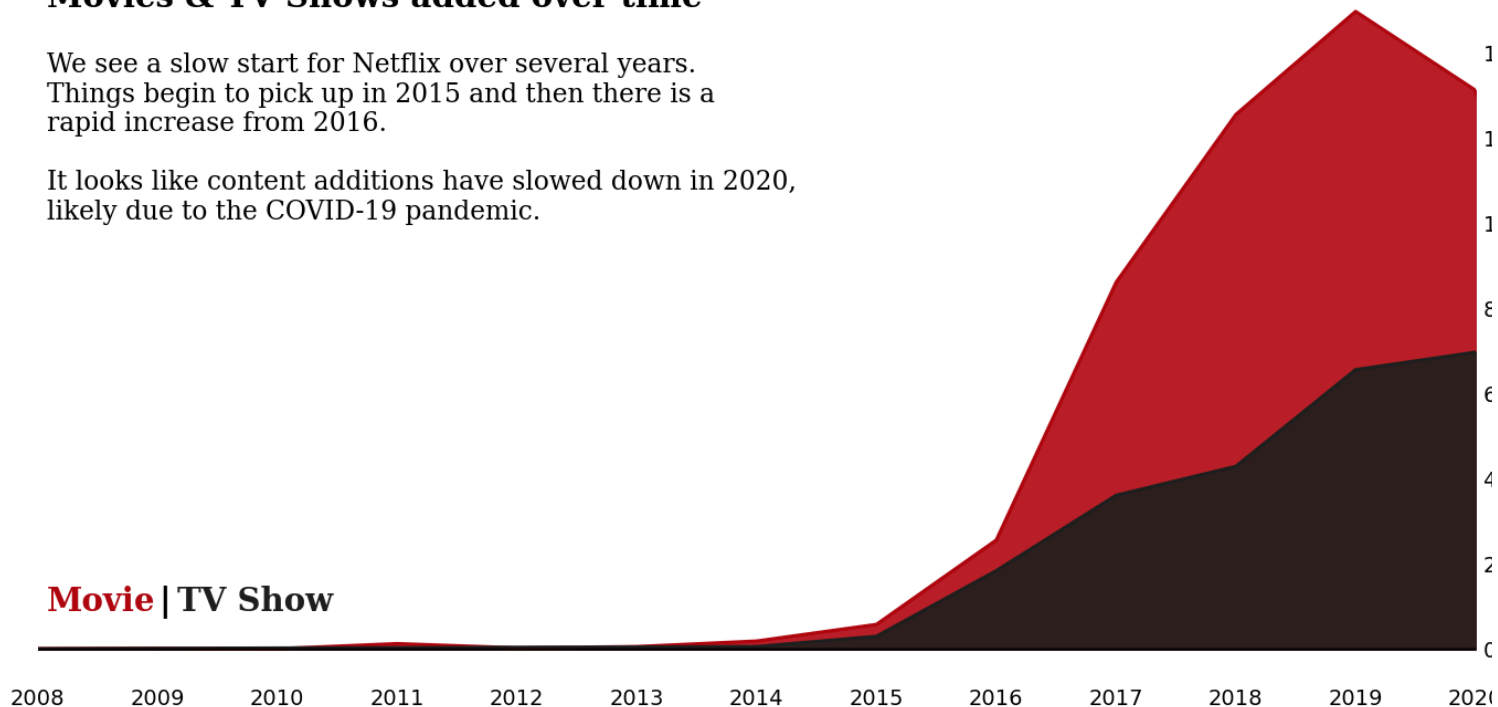
plt.show()

```

Movies & TV Shows added over time

We see a slow start for Netflix over several years.
Things begin to pick up in 2015 and then there is a
rapid increase from 2016.

It looks like content additions have slowed down in 2020,
likely due to the COVID-19 pandemic.



We can view the same plot, but as a cumulative total...

```

[19]: data_sub = df.groupby('type')['year_added'].value_counts().unstack().fillna(0).loc[['TV Show', 'Movie']].cumsum(axis=0).T

fig, ax = plt.subplots(1, 1, figsize=(12, 6))
color = ["#b20710", "#221f1f"]

for i, mtv in enumerate(df['type'].value_counts().index):
    mtv_rel = data_sub[mtv]

```

```

ax.fill_between(mtv_rel.index, 0, mtv_rel, color=color[i], label=mtv,alpha=0.9)

ax.yaxis.tick_right()

ax.axhline(y = 0, color = 'black', linewidth = 1.3, alpha = .7)

#ax.set_ylim(0, 50)
#ax.legend(loc='upper left')
for s in ['top', 'right','bottom','left']:
    ax.spines[s].set_visible(False)

ax.grid(False)

ax.set_xlim(2008,2020)
plt.xticks(np.arange(2008, 2021, 1))

fig.text(0.13, 0.85, 'Movies & TV Shows added over time [Cumulative Total]', fontsize=15, fontweight='bold', fontfamily='serif')
fig.text(0.13, 0.58,
'''Netflix peak global content amount was in 2019.

It appears that Netflix has focused more attention
on increasing Movie content that TV Shows.
Movies have increased much more dramatically
than TV shows.
''',
, fontsize=12, fontweight='light', fontfamily='serif')

fig.text(0.13,0.2,"Movie", fontweight="bold", fontfamily='serif', fontsize=15, color='#b20710')
fig.text(0.19,0.2,"|", fontweight="bold", fontfamily='serif', fontsize=15, color='black')
fig.text(0.2,0.2,"TV Show", fontweight="bold", fontfamily='serif', fontsize=15, color='#221f1f')

ax.tick_params(axis=u'both', which=u'both',length=0)

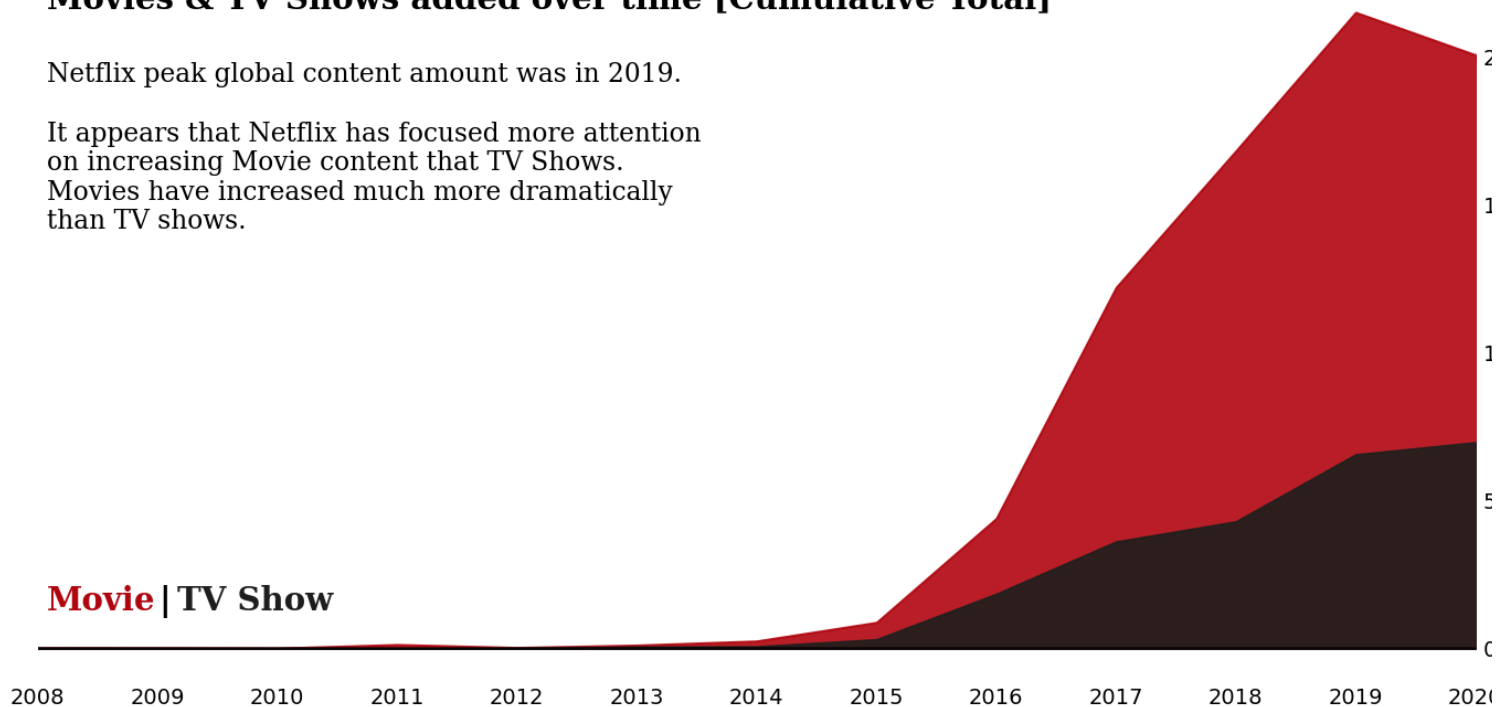
plt.show()

```

Movies & TV Shows added over time [Cumulative Total]

Netflix peak global content amount was in 2019.

It appears that Netflix has focused more attention on increasing Movie content that TV Shows. Movies have increased much more dramatically than TV shows.



Month-by-Month

We've seen how content has increased over the years, but are there certain months that, on average, tend to enjoy more content being added?

I'll show this in a couple of ways - a cumulative year view, and also as a radial plot...

```

[20]: month_order = ['January',
'February',
'March',
'April',
'May',

```

```

'June',
'July',
'August',
'September',
'October',
'November',
'December']

df['month_name_added'] = pd.Categorical(df['month_name_added'], categories=month_order, ordered=True)

```

[21]:

```

data_sub = df.groupby('type')['month_name_added'].value_counts().unstack().fillna(0).loc[['TV Show', 'Movie']].cumsum(axis=0).T

fig, ax = plt.subplots(1, 1, figsize=(12, 6))
color = ["#b20710", "#221f1f"]

for i, mtv in enumerate(df['type'].value_counts().index):
    mtv_rel = data_sub[mtv]
    ax.fill_between(mtv_rel.index, 0, mtv_rel, color=color[i], label=mtv, alpha=0.9)

ax.yaxis.tick_right()

ax.axhline(y = 0, color = 'black', linewidth = 1.3, alpha = .4)

#ax.set_ylim(0, 50)
#ax.legend(loc='upper left')
for s in ['top', 'right', 'bottom', 'left']:
    ax.spines[s].set_visible(False)

ax.grid(False)
ax.set_xticklabels(data_sub.index, fontfamily='serif', rotation=0)
ax.margins(x=0) # remove white spaces next to margins

#ax.set_xlim(2008,2020)
#plt.xticks(np.arange(2008, 2021, 1))

fig.text(0.13, 0.95, 'Content added by month [Cumulative Total]', fontsize=15, fontweight='bold', fontfamily='serif')
fig.text(0.13, 0.905,
"The end & beginnings of each year seem to be Netflix's preference for adding content."
, fontsize=12, fontweight='light', fontfamily='serif')

fig.text(0.13,0.855,"Movie", fontweight="bold", fontfamily='serif', fontsize=15, color='#b20710')
fig.text(0.19,0.855,"|", fontweight="bold", fontfamily='serif', fontsize=15, color='black')
fig.text(0.2,0.855,"TV Show", fontweight="bold", fontfamily='serif', fontsize=15, color='#221f1f')

ax.tick_params(axis=u'both', which=u'both',length=0)

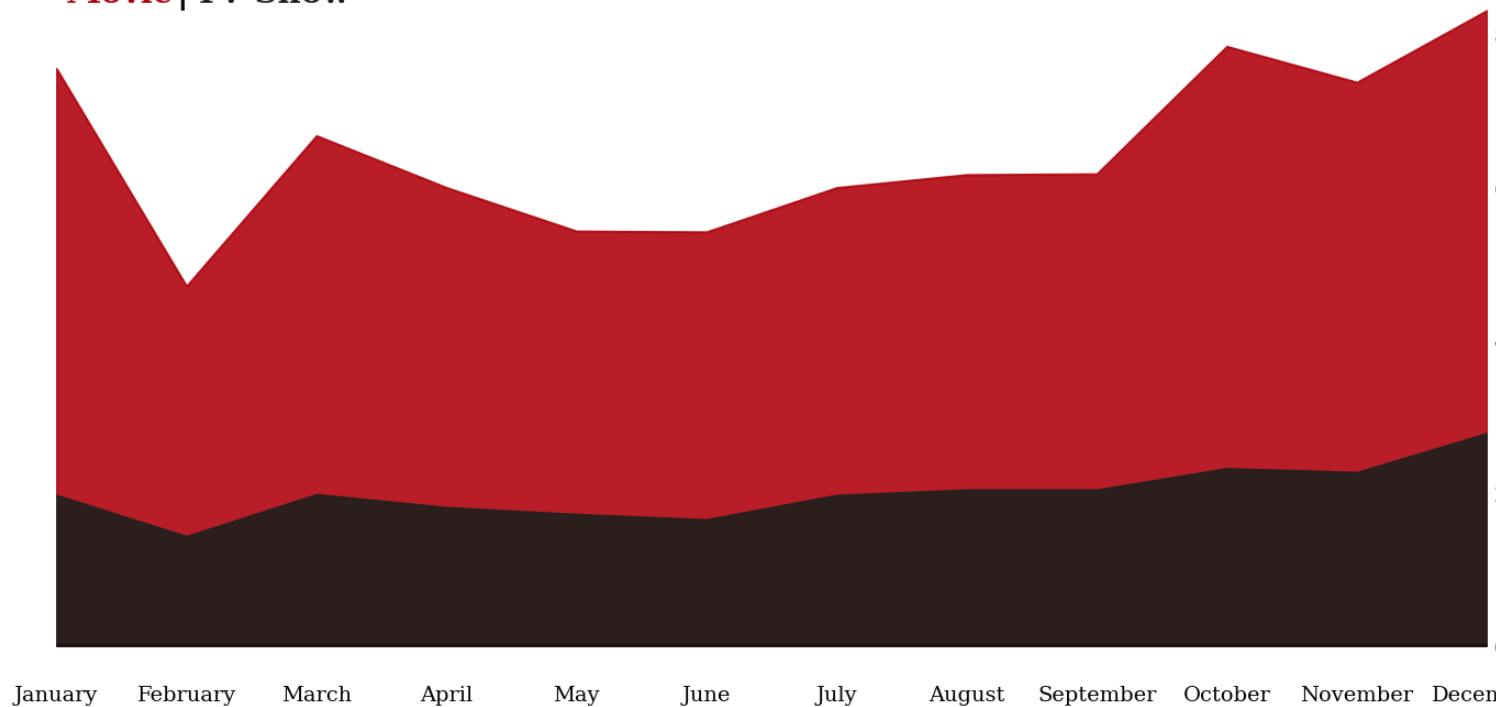
plt.show()

```

Content added by month [Cumulative Total]

The end & beginnings of each year seem to be Netflix's preference for adding content.

Movie | TV Show



What about a more interesting way to view how content is added across the year?

Sometimes visualizations should be eye-catching & attention grabbing - I think this visual achieves that, even if it isn't the most precise.

By highlighting certain months, the reader's eye is drawn exactly where we want it.

```
[22]: data_sub2 = data_sub

data_sub2['Value'] = data_sub2['Movie'] + data_sub2['TV Show']
data_sub2 = data_sub2.reset_index()

df_polar = data_sub2.sort_values(by='month_name_added', ascending=False)

color_map = ['#221f1f' for _ in range(12)]
color_map[0] = color_map[11] = '#b20710' # color highlight

# initialize the figure
plt.figure(figsize=(8,8))
ax = plt.subplot(111, polar=True)
plt.axis('off')

# Constants = parameters controlling the plot layout:
upperLimit = 30
lowerLimit = 1
labelPadding = 30

# Compute max and min in the dataset
max = df_polar['Value'].max()

# Let's compute heights: they are a conversion of each item value in those new coordinates
# In our example, 0 in the dataset will be converted to the lowerLimit (10)
# The maximum will be converted to the upperLimit (100)
slope = (max - lowerLimit) / max
heights = slope * df_polar.Value + lowerLimit

# Compute the width of each bar. In total we have 2*Pi = 360°
width = 2*np.pi / len(df_polar.index)

# Compute the angle each bar is centered on:
indexes = list(range(1, len(df_polar.index)+1))
angles = [element * width for element in indexes]
angles

# Draw bars
bars = ax.bar(
    x=angles,
```

```

height=heights,
width=width,
bottom=lowerLimit,
linewidth=2,
edgecolor="white",
color=color_map,alpha=0.8
)

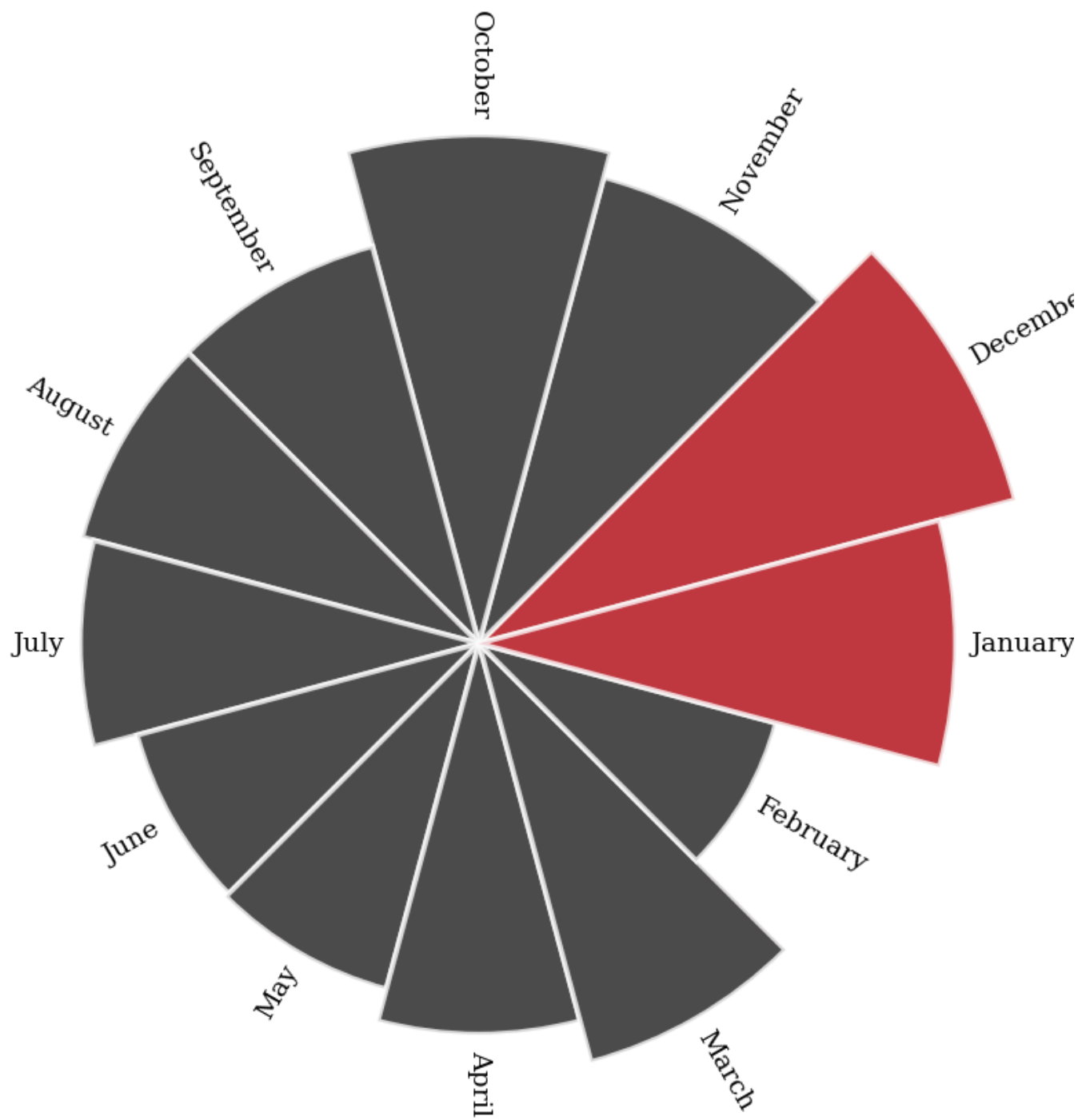
# Add labels
for bar, angle, height, label in zip(bars,angles, heights, df_polar["month_name_added"]):

    # Labels are rotated. Rotation must be specified in degrees :(
    rotation = np.rad2deg(angle)

    # Flip some labels upside down
    alignment = ""
    if angle >= np.pi/2 and angle < 3*np.pi/2:
        alignment = "right"
        rotation = rotation + 180
    else:
        alignment = "left"

    # Finally add the labels
    ax.text(
        x=angle,
        y=lowerLimit + bar.get_height() + labelPadding,
        s=label,
        ha=alignment, fontsize=10,fontfamily='serif',
        va='center',
        rotation=rotation,
        rotation_mode="anchor")

```



Yes, December & January are definitely the best months for new content. Maybe Netflix knows that people have a lot of time off from work over this period and that it is a good time to reel people in. February is the worst - why might this be? Ideas welcomed!

Movie Genres

Let's now explore movie genres a little...

```
[23]: # Genres
from sklearn.preprocessing import MultiLabelBinarizer

import matplotlib.colors

# Custom colour map based on Netflix palette
cmap = matplotlib.colors.LinearSegmentedColormap.from_list("", ['#221f1f', '#b20710', '#f5f5f1'])
```

```
def genre_heatmap(df, title):
    df['genre'] = df['listed_in'].apply(lambda x : x.replace(' ','').replace(', ','').split(','))
    Types = []
    for i in df['genre']: Types += i
    Types = set(Types)
    print("There are {} types in the Netflix {} Dataset".format(len(Types),title))
    test = df['genre']
    mlb = MultiLabelBinarizer()
    res = pd.DataFrame(mlb.fit_transform(test), columns=mlb.classes_, index=test.index)
    corr = res.corr()
    mask = np.zeros_like(corr, dtype=np.bool)
    mask[np.triu_indices_from(mask)] = True
    fig, ax = plt.subplots(figsize=(10, 7))
    fig.text(.54,.88,'Genre correlation', fontfamily='serif',fontweight='bold',fontsize=15)
    fig.text(.75,.665,
    '''
        It is interesting that Independant Movies
        tend to be Dramas.

        Another observation is that
        Internatinal Movies are rarely
        in the Children's genre.
    ''', fontfamily='serif',fontsize=12,ha='right')
    pl = sns.heatmap(corr, mask=mask, cmap=cmap, vmax=.3, vmin=-.3, center=0, square=True, linewidths=2.5)

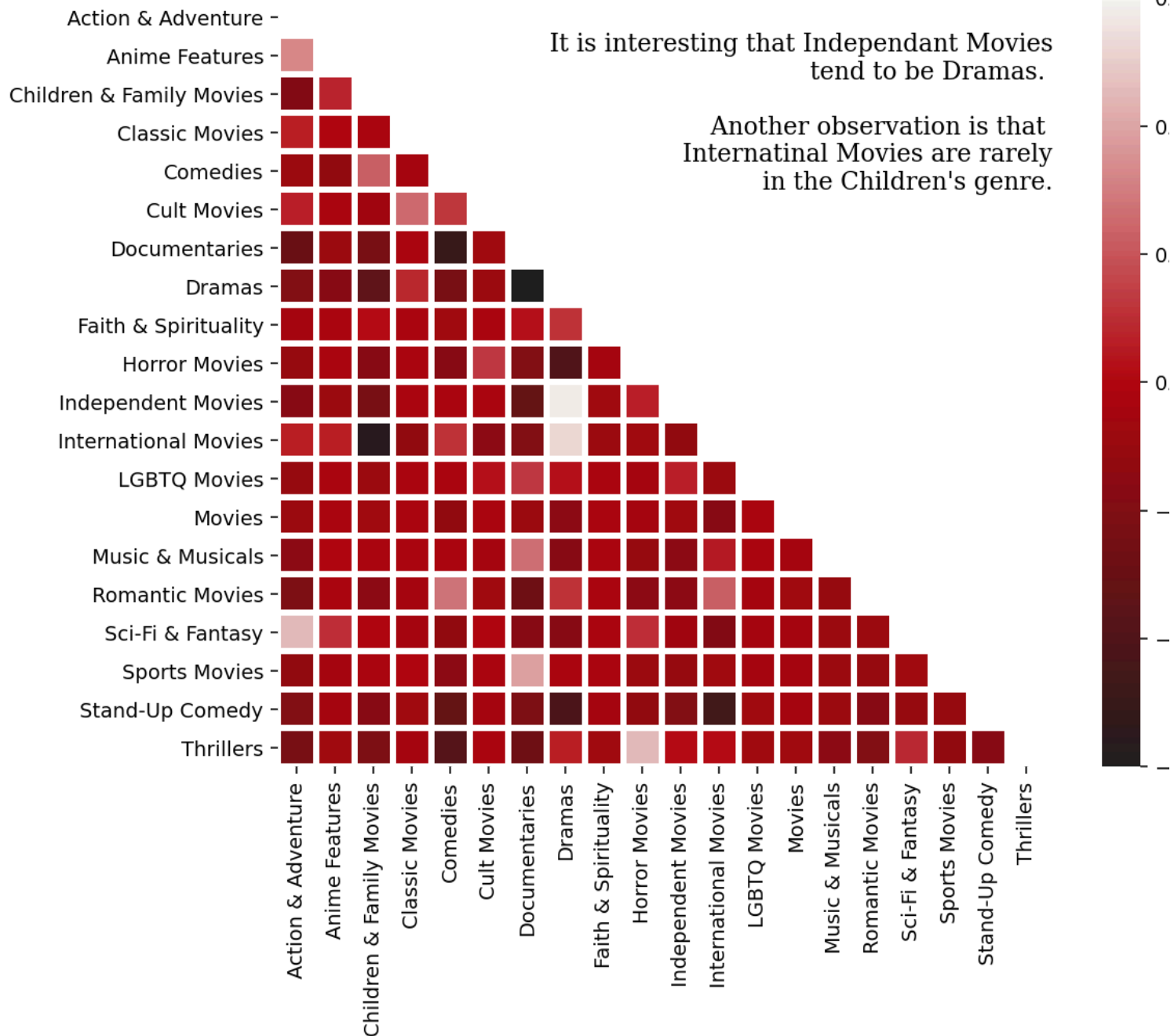
    plt.show()
```

```
[24]: df_tv = df[df["type"] == "TV Show"]
df_movies = df[df["type"] == "Movie"]
```

```
genre_heatmap(df_movies, 'Movie')
plt.show()
```

There are 20 types in the Netflix Movie Dataset

Genre correlation



[25]:

```
data = df.groupby('first_country')[['first_country', 'count']].sum().sort_values(by='count', ascending=False).reset_index()[:10]
data = data['first_country']

df_heatmap = df.loc[df['first_country'].isin(data)]
```

[26]:

```
df_heatmap = pd.crosstab(df_heatmap['first_country'], df_heatmap['target_ages'], normalize = "index").T
```

Target Ages

Does Netflix uniformly target certain demographics? Or does this vary by country?

[27]:

```
fig, ax = plt.subplots(1, 1, figsize=(12, 12))

country_order2 = ['USA', 'India', 'UK', 'Canada', 'Japan', 'France', 'S. Korea', 'Spain',
                  'Mexico', 'Turkey']

age_order = ['Kids', 'Older Kids', 'Teens', 'Adults']

sns.heatmap(df_heatmap.loc[age_order, country_order2], cmap=cmap, square=True, linewidth=2.5, cbar=False,
```

```

annot=True,fmt='1.0%',vmax=.6,vmin=0.05,ax=ax,annot_kws={"fontsize":12})

ax.spines['top'].set_visible(True)

fig.text(.99, .725, 'Target ages proportion of total content by country', fontweight='bold', fontfamily='serif', fontsize=15,ha='right')
fig.text(0.99, 0.7, 'Here we see interesting differences between countries. Most shows in India are targeted to teens, for instance.',ha='right', fontsize=12,fontfamily='serif')

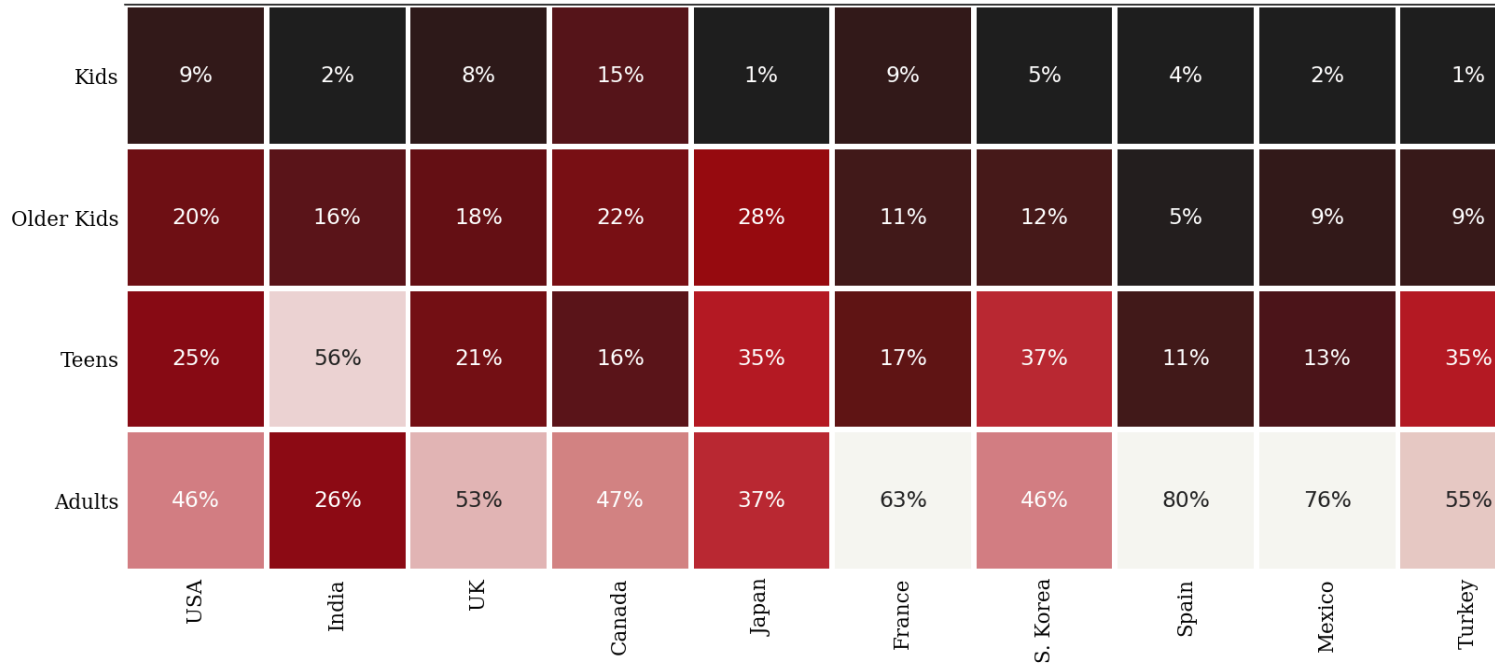
ax.set_yticklabels(ax.get_yticklabels(), fontfamily='serif', rotation = 0, fontsize=11)
ax.set_xticklabels(ax.get_xticklabels(), fontfamily='serif', rotation=90, fontsize=11)

ax.set_ylabel('')
ax.set_xlabel('')
ax.tick_params(axis=u'both', which=u'both',length=0)
plt.tight_layout()
plt.show()

```

Target ages proportion of total content by coun

Here we see interesting differences between countries. Most shows in India are targeted to teens, for insta



Very interesting results.

It is also interesting to note similarities between culturally similar countries - the US & UK are closey aligned with their Netflix target ages, yet vastly different to say, India or Japan!

Let's have a quick look at the lag between when content is released and when it is added on Netfli

Spain looks to have a lot of new content. Great for them!

```

[28]: # Data

df_movies
df_tv

### Relevant groupings

data = df_movies.groupby('first_country')[['first_country', 'count']].sum().sort_values(by='count',ascending=False).reset_index()[:10]
data = data['first_country']
df_loli = df_movies.loc[df_movies['first_country'].isin(data)]

loli = df_loli.groupby('first_country')['release_year', 'year_added'].mean().round()

# Reorder it following the values of the first value
ordered_df = loli.sort_values(by='release_year')

ordered_df_rev = loli.sort_values(by='release_year',ascending=False)

my_range=range(1,len(loli.index)+1)

fig, ax = plt.subplots(1, 1, figsize=(7, 5))

```

```

fig.text(0.13, 0.9, 'How old are the movies? [Average]', fontsize=15, fontweight='bold', fontfamily='serif')
plt.hlines(y=my_range, xmin=ordered_df['release_year'], xmax=ordered_df['year_added'], color='grey', alpha=0.4)
plt.scatter(ordered_df['release_year'], my_range, color='#221f1f',s=100, alpha=0.9, label='Average release date')
plt.scatter(ordered_df['year_added'], my_range, color='#b20710',s=100, alpha=0.9, label='Average added date')
plt.legend()

for s in ['top', 'left', 'right', 'bottom']:
    ax.spines[s].set_visible(False)

# Removes the tick marks but keeps the labels
ax.tick_params(axis=u'both', which=u'both',length=0)
# Move Y axis to the right side
ax.yaxis.tick_right()

plt.yticks(my_range, ordered_df.index)
plt.yticks(fontname = "serif",fontsize=12)

# Custome legend
fig.text(0.19,0.175,"Released", fontweight="bold", fontfamily='serif', fontsize=12, color='#221f1f')
fig.text(0.76,0.175,"Added", fontweight="bold", fontfamily='serif', fontsize=12, color='#b20710')

fig.text(0.13, 0.46,
'''The average gap between when
content is released, and when it
is then added on Netflix varies
by country.

In Spain, Netflix appears to be
dominated by newer movies
whereas Egypt & India have
an older average movie.
''',
, fontsize=12, fontweight='light', fontfamily='serif')

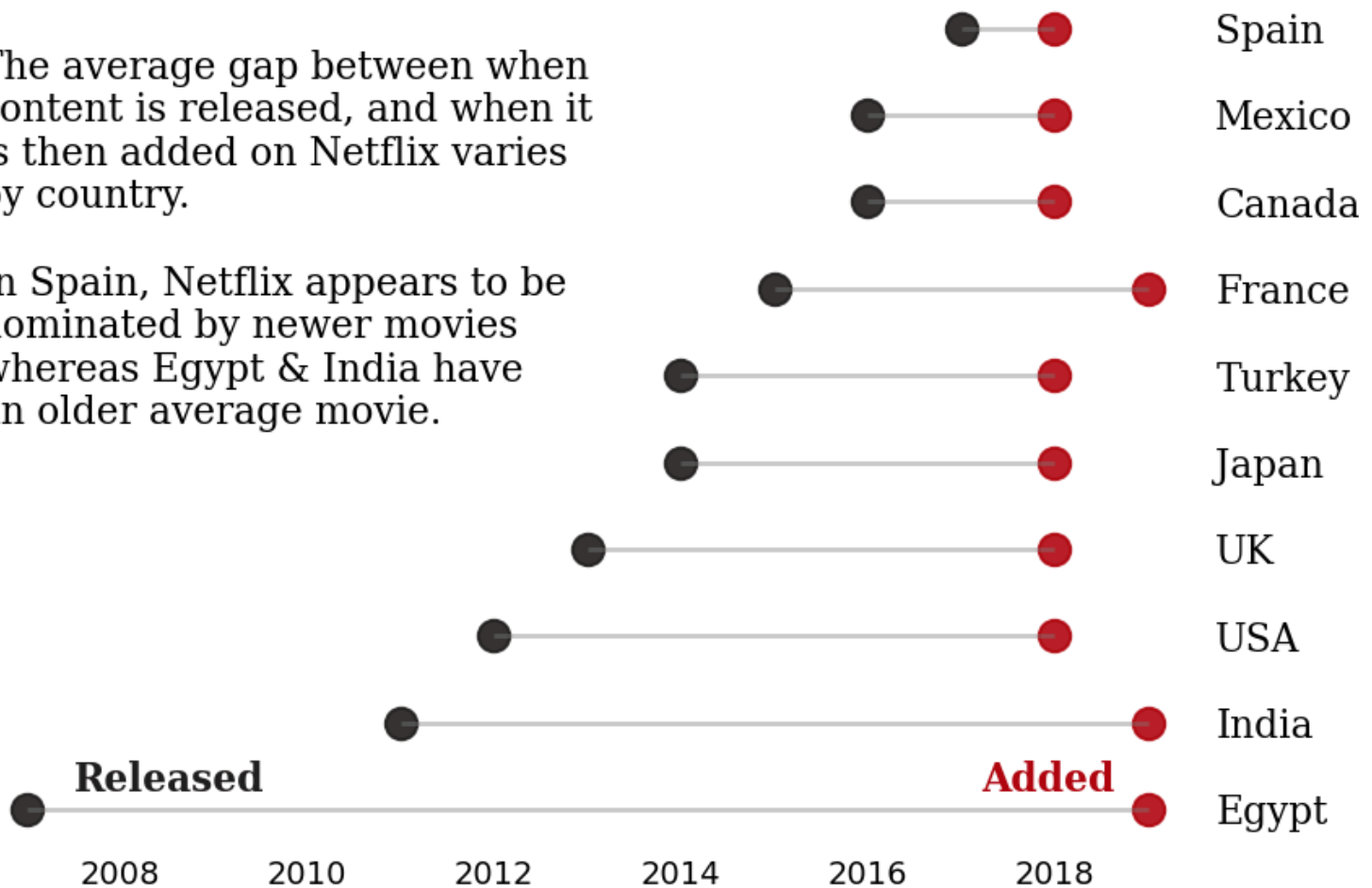
#plt.xlabel('Year')
#plt.ylabel('Country')
plt.show()

```

How old are the movies? [Average]

The average gap between when content is released, and when it is then added on Netflix varies by country.

In Spain, Netflix appears to be dominated by newer movies whereas Egypt & India have an older average movie.



What about TV shows...

```
[29]: data = df_tv.groupby('first_country')[['first_country', 'count']].sum().sort_values(by='count', ascending=False).reset_index()[0:10]
data = data['first_country']
df_loli = df_tv.loc[df_tv['first_country'].isin(data)]

loli = df_loli.groupby('first_country')['release_year', 'year_added'].mean().round()

# Reorder it following the values of the first value:
ordered_df = loli.sort_values(by='release_year')

ordered_df_rev = loli.sort_values(by='release_year', ascending=False)

my_range = range(1, len(loli.index)+1)

fig, ax = plt.subplots(1, 1, figsize=(7, 5))

fig.text(0.13, 0.9, 'How old are the TV shows? [Average]', fontsize=15, fontweight='bold', fontfamily='serif')
plt.hlines(y=my_range, xmin=ordered_df['release_year'], xmax=ordered_df['year_added'], color='grey', alpha=0.4)
plt.scatter(ordered_df['release_year'], my_range, color='#221f1f', s=100, alpha=0.9, label='Average release date')
plt.scatter(ordered_df['year_added'], my_range, color='#b20710', s=100, alpha=0.9, label='Average added date')
#plt.legend()

for s in ['top', 'left', 'right', 'bottom']:
    ax.spines[s].set_visible(False)

ax.yaxis.tick_right()
plt.yticks(my_range, ordered_df.index)
plt.yticks(fontname="serif", fontsize=12)

fig.text(0.19, 0.175, "Released", fontweight="bold", fontfamily='serif', fontsize=12, color='#221f1f')

fig.text(0.47, 0.175, "Added", fontweight="bold", fontfamily='serif', fontsize=12, color='#b20710')

fig.text(0.13, 0.42,
```

```
'''The gap for TV shows seems
more regular than for movies.

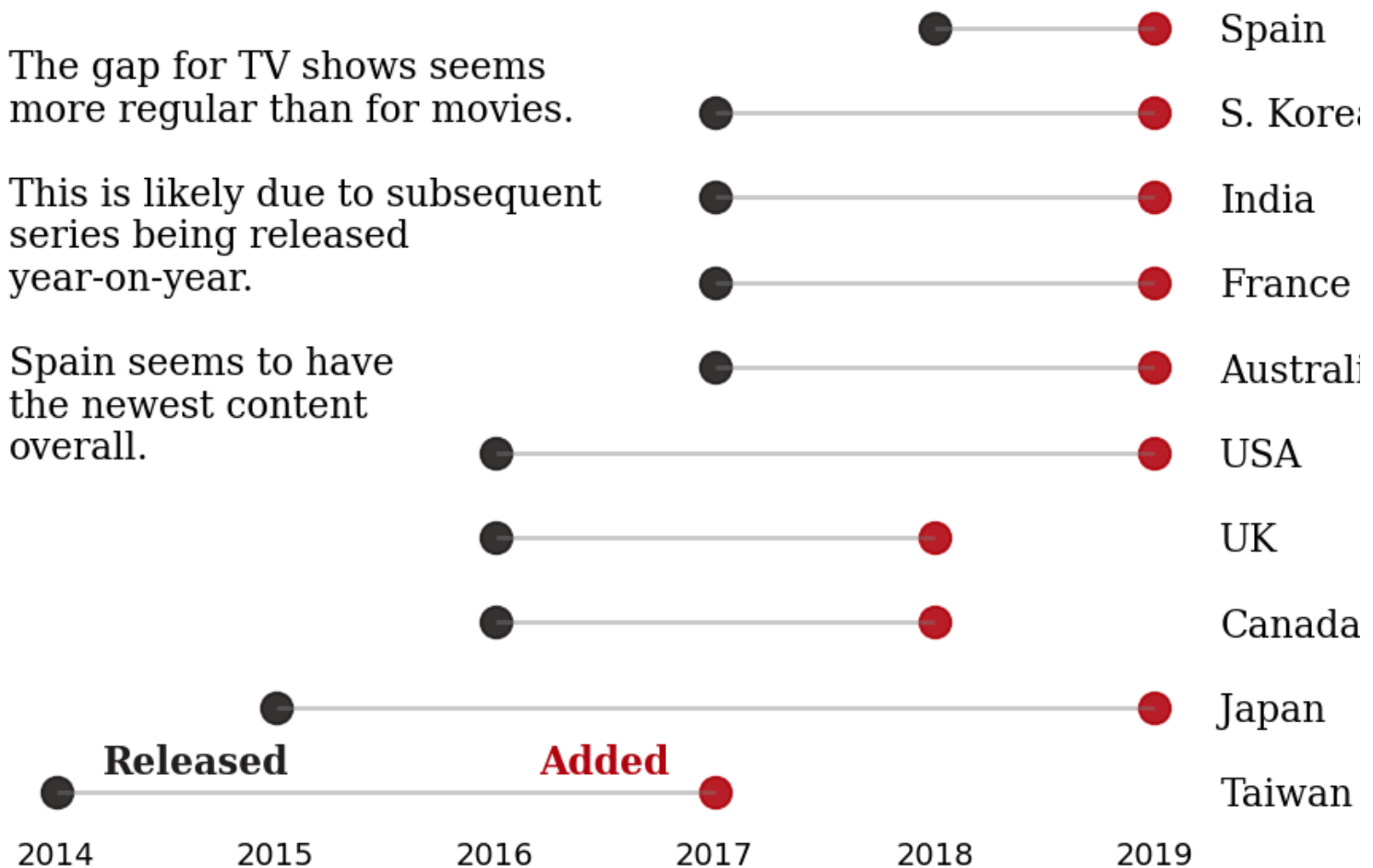
This is likely due to subsequent
series being released
year-on-year.

Spain seems to have
the newest content
overall.
'''

, fontsize=12, fontweight='light', fontfamily='serif')

ax.tick_params(axis=u'both', which=u'both',length=0)
#plt.xlabel('Value of the variables')
#plt.ylabel('Group')
plt.show()
```

How old are the TV shows? [Average]



[30]:

```
us_ind = df[(df['first_country'] == 'USA') | (df['first_country'] == 'India' )]

data_sub = df.groupby('first_country')['year_added'].value_counts().unstack().fillna(0).loc[['USA', 'India']].cumsum(axis=0).T
```

USA & India

As the two largest content countries, it might be fun to compare the two

[31]:

```
fig, ax = plt.subplots(1, 1, figsize=(12, 6))
color = ['221f1f', '#b20710', '#f5f5f1']

for i, hs in enumerate(us_ind['first_country'].value_counts().index):
    hs_built = us_ind[us_ind['first_country']==hs]['year_added'].value_counts().sort_index()
    ax.plot(hs_built.index, hs_built, color=color[i], label=hs)
    #ax.fill_between(hs_built.index, 0, hs_built, color=color[i], alpha=0.4)
    ax.fill_between(hs_built.index, 0, hs_built, color=color[i], label=hs)
```

```

ax.set_ylim(0, 1000)
#ax.legend(loc='upper left')
for s in ['top', 'right']:
    ax.spines[s].set_visible(False)

ax.yaxis.tick_right()

ax.axhline(y = 0, color = 'black', linewidth = 1.3, alpha = .4)

#ax.set_ylim(0, 50)
#ax.legend(loc='upper left')
for s in ['top', 'right', 'bottom', 'left']:
    ax.spines[s].set_visible(False)

ax.grid(False)
ax.set_xticklabels(data_sub.index, fontfamily='serif', rotation=0)
ax.margins(x=0) # remove white spaces next to margins

ax.set_xlim(2008, 2020)
plt.xticks(np.arange(2008, 2021, 1))

fig.text(0.13, 0.85, 'USA vs. India: When was content added?', fontsize=15, fontweight='bold', fontfamily='serif')
fig.text(0.13, 0.58,
'''
We know from our work above that Netflix is dominated by the USA & India.
It would also be reasonable to assume that, since Netflix is an American
company, Netflix increased content first in the USA, before
other nations.

That is exactly what we see here; a slow and then rapid
increase in content for the USA, followed by Netflix
being launched to the Indian market in 2016.'''
, fontsize=12, fontweight='light', fontfamily='serif')

fig.text(0.13, 0.15, "India", fontweight="bold", fontfamily='serif', fontsize=15, color='#b20710')
fig.text(0.188, 0.15, "|", fontweight="bold", fontfamily='serif', fontsize=15, color='black')
fig.text(0.198, 0.15, "USA", fontweight="bold", fontfamily='serif', fontsize=15, color='#221f1f')

ax.tick_params(axis=u'both', which=u'both',length=0)

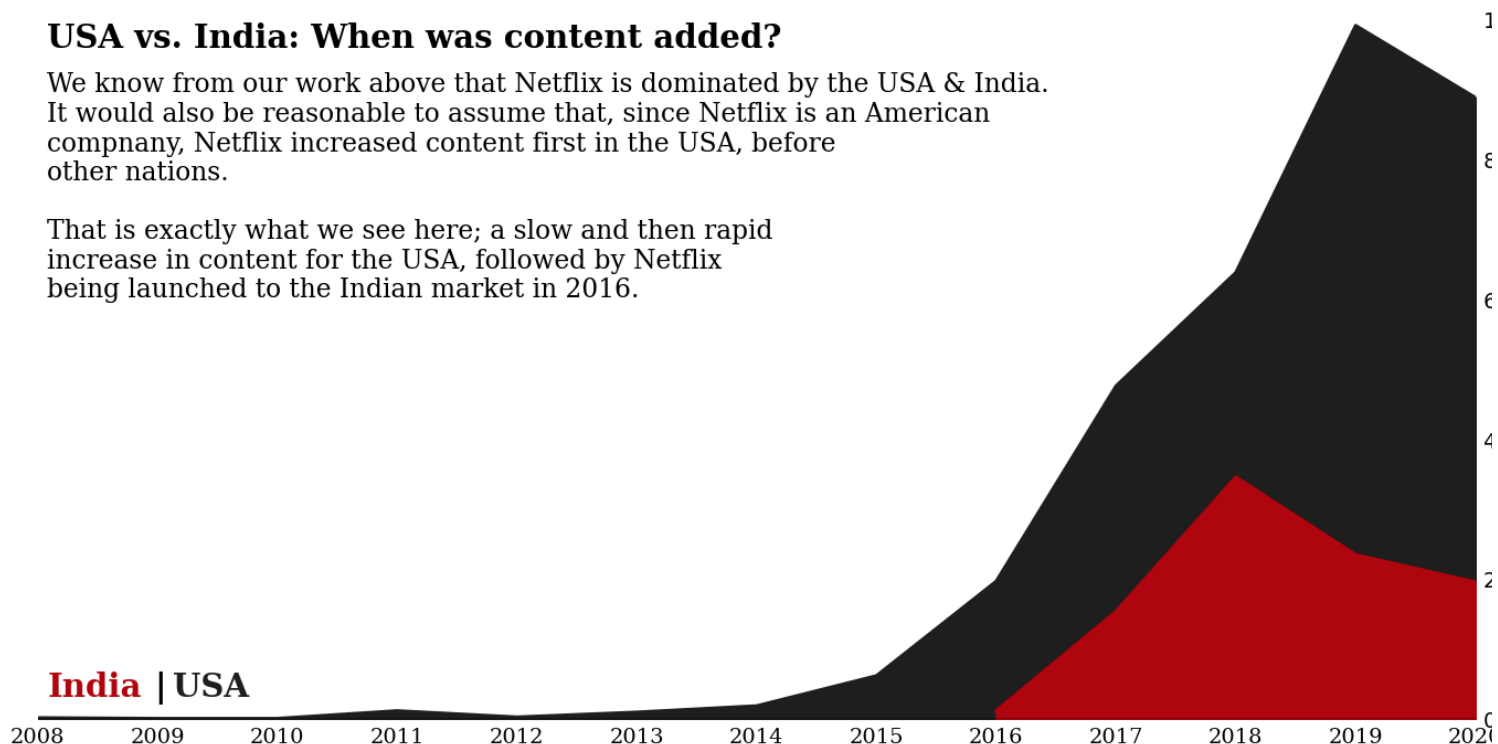
plt.show()

```

USA vs. India: When was content added?

We know from our work above that Netflix is dominated by the USA & India. It would also be reasonable to assume that, since Netflix is an American company, Netflix increased content first in the USA, before other nations.

That is exactly what we see here; a slow and then rapid increase in content for the USA, followed by Netflix being launched to the Indian market in 2016.



```

[32]: us_ind = df[(df['first_country'] == 'USA') | (df['first_country'] == 'India')]

data_sub = df.groupby('first_country')['year_added'].value_counts().unstack().fillna(0).loc[['USA', 'India']].cumsum(axis=0).T
data_sub.insert(0, "base", np.zeros(len(data_sub)))

```

```
data_sub = data_sub.add(-us_ind['year_added'].value_counts()/2, axis=0)
```

So the USA dominates. But is there a plot that can convey this in another way?

```
[33]: fig, ax = plt.subplots(1, 1, figsize=(14, 6))
color = ['#b20710', '#221f1f'][:-1]
hs_list = data_sub.columns
hs_built = data_sub[hs]

for i, hs in enumerate(hs_list):
    if i == 0 : continue
    ax.fill_between(hs_built.index, data_sub.iloc[:,i-1], data_sub.iloc[:,i], color=color[i-1])

for s in ['top', 'right', 'bottom', 'left']:
    ax.spines[s].set_visible(False)
ax.set_axisbelow(True)
ax.set_yticks([])
#ax.legend(loc='upper left')
ax.grid(False)

fig.text(0.16, 0.76, 'USA vs. India: Stream graph of new content added', fontsize=15, fontweight='bold', fontfamily='serif')
fig.text(0.16, 0.575,
'''
Seeing the data displayed like this helps
us to realise just how much content is added in the USA.
Remember, India has the second largest amount of
content yet is dwarfed by the USA.'''
, fontsize=12, fontweight='light', fontfamily='serif')

fig.text(0.16,0.41,"India", fontweight="bold", fontfamily='serif', fontsize=15, color='#b20710')
fig.text(0.208,0.41,"|", fontweight="bold", fontfamily='serif', fontsize=15, color='black')
fig.text(0.218,0.41,"USA", fontweight="bold", fontfamily='serif', fontsize=15, color='#221f1f')

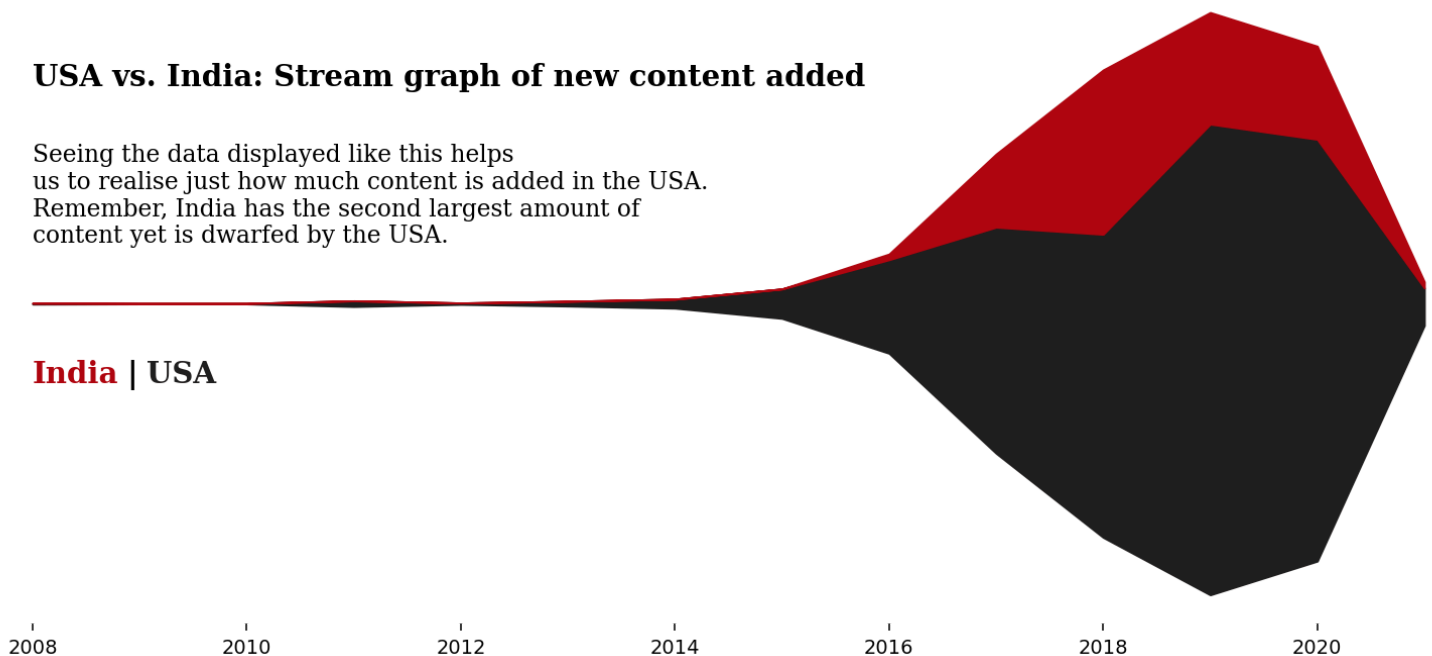
ax.tick_params(axis='y', which='both',length=0)

plt.show()
```

USA vs. India: Stream graph of new content added

Seeing the data displayed like this helps
us to realise just how much content is added in the USA.
Remember, India has the second largest amount of
content yet is dwarfed by the USA.

India | USA



Lastly, we can view a wordcloud to get an overview of Netflix titles

It is interesting to note that many films share the same key words in their titles.

```
[34]: from wordcloud import WordCloud
import random
from PIL import Image
import matplotlib
```

```
# Custom colour map based on Netflix palette
cmap = matplotlib.colors.LinearSegmentedColormap.from_list("", ['#221f1f', '#b20710'])

text = str(list(df['title'])).replace(',', '').replace('[', '').replace('"', '').replace(']', '').replace('.', '')

mask = np.array(Image.open('../input/netflix-icon-new/f6974e017d3f6196c4cbe284ee3eaf4e.png'))

wordcloud = WordCloud(background_color = 'white', width = 500, height = 200, colormap=cmap, max_words = 150, mask = mask).generate(text)

Project completed - key visualizations and analysis are retained above.
plt.figure(figsize=(5,5))
plt.imshow(wordcloud, interpolation = 'bilinear')
plt.axis('off')
plt.tight_layout(pad=0)
plt.show()
```



Thanks for reading!

I hope you enjoyed my visuals

Have a great day

[]: